28 A particle of mass m in a well of width L has ground-state energy E.

A second particle has mass $2m$ and is confined in a well of width $2L$.

Its ground-state energy is:

A $E/2$

B $E/4$

C $E/8$

D $E/16$

Ans: C

Using: $E_n = \dfrac{h^2}{8mL^2}n^2$, we have

first particle ground state energy, $E = \dfrac{h^2}{8mL^2}(1)^2 = \dfrac{h^2}{8mL^2}$,

second particle ground state energy, $E' = \dfrac{h^2}{8(2m)(2L)^2}(1)^2 = \dfrac{1}{8}\left(\dfrac{h^2}{8mL^2}\right) = \dfrac{1}{8}E$